

Code-based cryptography over the Lee Metric

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Proposal	Type	pk Low (kB)	pk High (kB)	Security
<i>Public-Key Encryption (PKE)/Key Encapsulation Mechanism (KEM)</i>				
Classic McEliece	Code	261,120	1,357,824	IND-CCA2
CRYSTALS-KYBER	Lattice	1.632	3.168	IND-CCA
NTRU	Lattice	0.931	1.230	IND-CCA
SABER	Lattice	0.672	1.312	IND-CCA
<i>Digital Signature Algorithm (DSA)</i>				
CRYSTALS-DILITHIUM	Lattice	1.312	2.592	SUF-CMA
FALCON	Lattice	0.897	1.793	EUFCMA
Rainbow	Multivariate	157.8	1885.4	EUFCMA

Table: Public key sizes of the National Institute of Standards and Technology (NIST) finalists [1].

Outline

- 1 Coding Theory
- 2 Code-based Cryptography
- 3 Statistical Decoding ($\mathbb{F}_q$ /Hamming)
- 4 SD ($\mathbb{Z}_{p^m}$ /Lee)
- 5 Comparison
- 6 Discussion

Coding Theory

Error Correcting Code

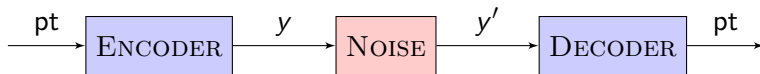


Figure: Error-correcting code.

We will use a specific type of error-correcting code called: Linear code

Definition ($\mathbb{F}_q$ -Code)

A $[n, k]_q$ -code $\mathcal{C}$ is a $\mathbb{F}_q$ -subspace of $\mathbb{F}_q^n$ of dimension k .

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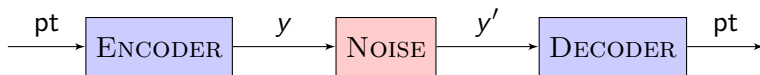


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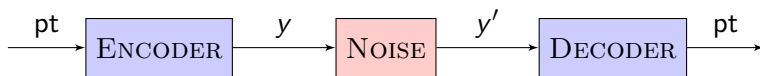


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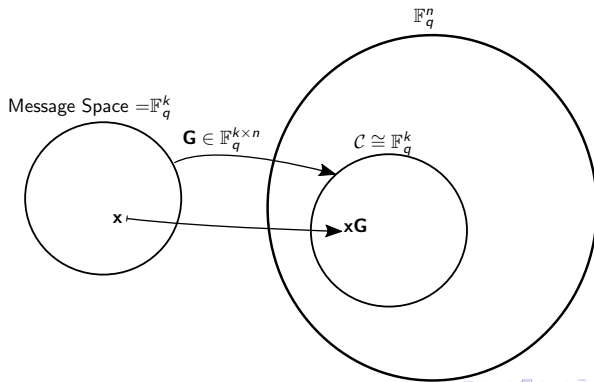
Notation

For the sake of simplicity, we will write $F = \mathbb{F}_q$ and $R = \mathbb{Z}_{p^m}$.

Generator Matrix

Definition (Generator Matrix)

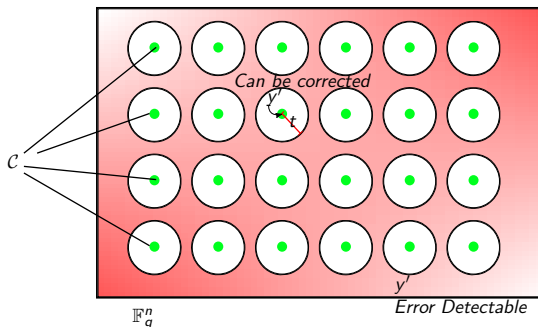
Given $\mathbf{G} \in F^{k \times n}$, if $\mathcal{C} = \{\vec{x}\mathbf{G} \mid \vec{x} \in F^k\}$, we say that $\mathbf{G}$ is a *generator matrix* of the code $\mathcal{C}$.



Error Correction Capability

Definition

We say that $\mathcal{C}$ has an error correction capability of t if it can correct elements that are at most t **far** from any $x \in \mathcal{C}$. We say that $\mathcal{C}$ is an $[n, k, 2t + 1]_q$ -code.



Metric

Hamming

$$d_H(\vec{x}, \vec{y}) = |\{i \in \llbracket n \rrbracket \mid x_i \neq y_i\}|$$

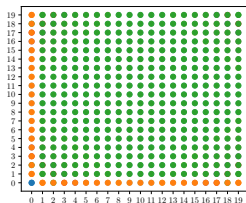
Lee

$$\begin{aligned} d_L(\vec{x}, \vec{y}) &= \sum_{i \in \llbracket n \rrbracket} \min(|x_i - y_i|, q - |x_i - y_i|) \\ &= \text{"} L_1\text{-distance that collapses"} \end{aligned}$$

Weight

Hamming

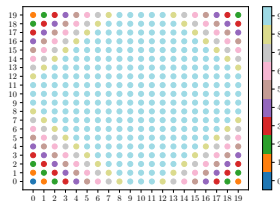
$$\text{wt}_H(\vec{x}) = |\{i \in \llbracket n \rrbracket \mid x_i \neq 0\}|$$



(a) Hamming weight.

Lee

$$\text{wt}_L(\vec{x}) = \sum_{i \in \llbracket n \rrbracket} \min(x_i, q - x_i)$$



(b) Lee weight.

Figure: Hamming and Lee weights of the elements of the ring $(\mathbb{Z}_{20})^2$.

Dual Code

Definition (Dual Code)

The dual code of $\mathcal{C}$, denoted by $\mathcal{C}^\perp$, is defined as:

$$\mathcal{C}^\perp = \{\vec{x} \in F^n \mid \forall \vec{y} \in \mathcal{C} \quad \vec{x}\vec{y}^\top = 0\}.$$

The Hardness of Decoding

Definition (Decoding problem)

Let $\vec{x} \in \mathcal{C}$. Given $\vec{x} + \vec{e}$, finds $\vec{x}$.

Theorem (Hardness of Decoding in the Hamming and Lee metric [2, 12])

*The problem of decoding a codeword generated by a random $\mathbb{F}_q$ -code (Hamming) or $\mathbb{Z}_m$ -code (Lee) is NP-hard in **average**.*

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- Obviously, the specific families of codes used in practice have an efficient decoding algorithm. Some examples of families of *Efficiently Decodable Codes*:
 - ▶ Hamming codes [6];
 - ▶ Hadamard codes;
 - ▶ Simplex codes;
 - ▶ Goppa codes [5];
- ★ **Binary Goppa codes.**

Post-Quantum Security

NP-hardness $\nrightarrow$ PQ-safe

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Generic Decoding Immune to **strong**
Fourier sampling [4].

Code-based Cryptography

McEliece Cryptosystem

GEN(1^λ)

1 : $\mathbf{G} \leftarrow \mathcal{F}$ = Binary Goppa Code
that can decode up to t errors.

2 : $\mathbf{P} \leftarrow P_n$

3 : $\mathbf{S} \leftarrow \text{GL}_k(F)$

4 : $\hat{\mathbf{G}} \leftarrow \mathbf{SGP}$

5 : **return** ($\text{pk} = (\hat{\mathbf{G}}, t)$, $\text{sk} = (\mathbf{S}, \mathbf{G}, \mathbf{P})$)
such that $\hat{\mathbf{G}}$ looks random.

ENC($\text{pk} = (\hat{\mathbf{G}}, t), \vec{x}$)

1 : $\vec{e} \leftarrow \{ \vec{e} \in F^n \mid \text{wt}_H(\vec{e}) = t \}$

2 : **return** $\vec{x}\hat{\mathbf{G}} + \vec{e}$

DEC($\text{sk} = (\mathbf{S}, \mathbf{G}, \mathbf{P}), \vec{y}$)

1 : $\vec{y}' \leftarrow \vec{y}\mathbf{P}^{-1}$

2 : $\vec{y}'' \leftarrow \mathcal{D}_{\mathbf{G}}(\vec{y}')$

3 : $\vec{x} = \vec{y}''\mathbf{S}^{-1}$; **return** $\vec{x}$
such that $\vec{y}' = \vec{x}\mathbf{SG} + \vec{e}\mathbf{P}^{-1}$.

Lee-McEliece Cryptosystem

GEN(1^λ)

1 : $\mathbf{G} \leftarrow \$ \mathcal{F}$

2 : $\mathbf{P} \leftarrow \$ P_n$

3 : $\mathbf{S}_1 \leftarrow \$ GL_{k_1}(R)$

$\vdots$

$\mathbf{S}_m \leftarrow \$ GL_{k_m}(R)$

4 : $\mathbf{S} \leftarrow \begin{pmatrix} \mathbf{S}_1 & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_2 & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{0} \\ \vdots & \vdots & & \vdots & \vdots & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \cdots & \mathbf{0} & \mathbf{S}_m \end{pmatrix}$

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- ISD gets worse on $\mathbb{Z}_m$ /Lee [11]...
- $\implies$ We can drastically reduce the public key size
- How about another statistical decoding? statistical decoding!
- How does statistical decoding adapt to $\mathbb{Z}_m$ /Lee.

Statistical Decoding ($\mathbb{F}_q$ /Hamming)

$$\text{Recall that } \underbrace{\left(\overbrace{\vec{x}}^{\in \mathcal{C}} + \overbrace{\vec{e}}^{\notin \mathcal{C}} \right)}_{\in \mathcal{C}^\perp} \underbrace{\vec{h}^\top}_{=0} = \vec{x}\vec{h} + \vec{e}\vec{h}^\top = \vec{e}\vec{h}^\top.$$
$$\begin{array}{c|cccccccccccc} \vec{h} & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ \vec{e} & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ \hline e_i h_i & 0 & + & 0 & + & 0 & + & 1 & + & 0 & + & 1 & + & 0 & + & 0 & + & 0 & + & 1 & 0 & 0 & \equiv 3 \equiv 1 \pmod{2} \end{array}$$

The Statistical Bias

Recall that $\underbrace{\left(\overbrace{\vec{x}}^{\in \mathcal{C}} + \overbrace{\vec{e}}^{\notin \mathcal{C}} \right)}_{\in \mathcal{C}^\perp} \vec{h}^\top = \underbrace{\vec{x}\vec{h}}_{=0} + \vec{e}\vec{h}^\top = \vec{e}\vec{h}^\top.$

Suppose we are working over $\mathbb{F}_2$. If $\vec{e}\vec{h}^\top = \sum_{i=1}^n e_i h_i \equiv 1 \pmod{2}$

- More generally, over $\mathbb{F}_q$, when $\vec{e}\vec{h}^\top \neq 0$, we say that we have an *odd error detection*

$\vec{h}$	0	0	0	1	0	1	0	0	1	1	0	0										
$\vec{e}$	0	0	1	1	0	1	0	0	0	1	0	0										
$e_i h_i$	0	+	0	+	0	+	1	+	0	+	1	+	0	+	0	+	0	+	1	0	0	$\equiv 3 \equiv 1 \pmod{2}$

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- $\implies$ Then there must be an *odd* number of i such that $e_i = h_i = 1$.

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$e_i h_i$	0	+	0	+	0	+	1	+	0	+	1	+	0	+	0	+	0	+	1	0	0	≡ 3 ≡ 1 (mod 2)

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- $\implies$ Then there must be an **odd** number of i such that $e_i = h_i = 1$.
- $\implies \exists i \quad e_i = h_i = 1$ (There is an error at a non-zero position of $\vec{y}$)
- Repeat for many $\vec{h}$ such that there is an odd error detection until it's clear which positions of $\vec{e}$ are ones.

$$\begin{array}{c|cccccccccccccccc} \vec{h} & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 \\ \vec{e} & 0 & 0 & 1 & 1 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ \hline e_i h_i & 0 & + & 0 & + & 0 & + & 1 & + & 0 & + & 1 & + & 0 & + & 0 & + & 0 & + & 1 & 0 & 0 & \equiv 3 \equiv 1 \pmod{2} \end{array}$$

Example of the Statistical Bias

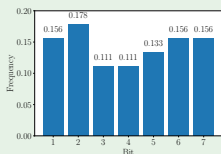
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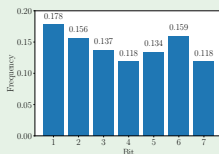
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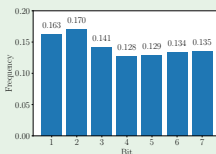
Let $\vec{e} = [1, 1, 0, 0, 0, 0, 0]$ and $\vec{v} = \sum_{\vec{h} \in \mathcal{H}_w} (\underbrace{\vec{y} \vec{h}^\top}_{\text{odd error detection}}) \vec{h}$.



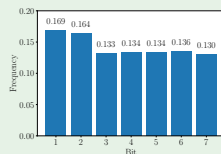
(a) $|\mathcal{H}_w| = 2 \cdot 10^1$.



(b) $|\mathcal{H}_w| = 2 \cdot 10^2$.



(c) $|\mathcal{H}_w| = 2 \cdot 10^3$.



(d) $|\mathcal{H}_w| = 2 \cdot 10^4$.

Figure: Frequencies of the counting vector $\vec{v}$ for different sizes of $\mathcal{H}_w$ (20, 200, 2000, 20000) with $w = 3$.

The Algorithm over $\mathbb{F}_q$ /Hamming [7, 8]

Definition

Let $\Pi: x \mapsto \mathbb{1}_{x \neq 0}$.

Finite Field Statistical Decoding($\mathcal{H}_w, \vec{y}$)

- 1 : // Where the outer sum and the scalar product are done over $\mathbb{Z}$
- 2 : // whereas the inner product is over F
- 3 : $\vec{v} \leftarrow \sum_{\vec{h} \in \mathcal{H}_w} (\Pi(\vec{y}\vec{h}^\top) \vec{h})$
- 4 : // Get the set of non-erroneous indices
- 5 : $\mathcal{I} \leftarrow \{\text{Indices of the } k \text{ smallest values of } \vec{v} \text{ such that } \mathbf{G}_{\mathcal{I}} \text{ is invertible}\}$
- 6 : // Invert the codeword on its information set
- 7 : **return** $\vec{y}_{\mathcal{I}}(\mathbf{G}_{\mathcal{I}})^{-1}$

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- We have to precompute $\mathcal{H}_w$, which is NP-hard to compute.

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- Is it still true over the Lee metric?

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- However, Niebuhr [8] has shown that over $\mathbb{F}_q$, statistical decoding outperforms ISD for q large enough.
- Is it still true over rings?
- Is it still true over the Lee metric?
- And if we combine rings and the Lee metric?

SD ($\mathbb{Z}_{p^m}$ /Lee)

Adapting the Algorithm

Both algorithms adapt nicely to $\mathbb{Z}_{p^m}$ embedded with the Lee metric [11].

Statistical Decoding over $\mathbb{Z}_{p^m}/\text{Lee}$

$\mathbb{Z}_{p^m}/\text{Lee}$ Statistical Decoding($\mathcal{H}_w, \vec{y}$)

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- The same with $\text{wt}_L(\vec{e}) = t$ instead of $\text{wt}_H(\vec{e}) = t$.
 - The algorithm mostly depends on $\text{wt}_H(\vec{e})$: the statistical bias is pretty much the same;
 - So an instance of Lee-SD decoding with $\text{wt}_L(\vec{e}) = t$ **reduces** to Hamming-SD with $\text{wt}_H(\vec{e})$.

Lee Statistical Decoding (t) $\approx$ Hamming Statistical Decoding (μ) over $\mathbb{Z}_m$

From now on we let $\mu = \mathbb{E}[\text{wt}_H(\vec{e}) \mid \text{wt}_L(\vec{e}) = t]$

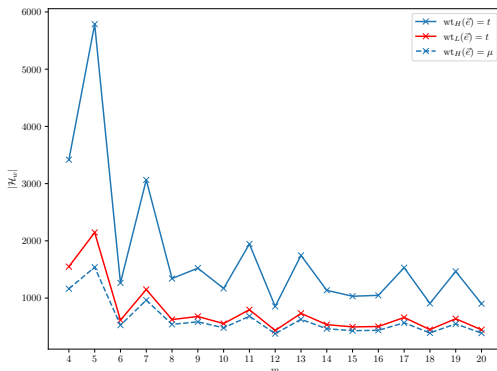


Figure: $|\mathcal{H}_w|$ to achieve 95% probability of success with $t = 7$, $w = 9$ and $n = 30$.

Statistical Decoding over $\mathbb{Z}_m$ /Hamming

$ \mathcal{H}_w $	Ambient Space	Success Frequency	$ \mathcal{H}_w $	Ambient Space	Success Frequency
5	$\mathbb{Z}_4$	0.584	5	$\mathbb{F}_4$	0.530
5	$\mathbb{Z}_8$	0.627	5	$\mathbb{F}_8$	0.598
5	$\mathbb{Z}_{16}$	0.622	5	$\mathbb{F}_{16}$	0.618
$5 \cdot 10$	$\mathbb{Z}_4$	0.785	$5 \cdot 10$	$\mathbb{F}_4$	0.744
$5 \cdot 10$	$\mathbb{Z}_8$	0.828	$5 \cdot 10$	$\mathbb{F}_8$	0.873
$5 \cdot 10$	$\mathbb{Z}_{16}$	0.863	$5 \cdot 10$	$\mathbb{F}_{16}$	0.896
$5 \cdot 10^2$	$\mathbb{Z}_4$	0.978	$5 \cdot 10^2$	$\mathbb{F}_4$	0.873
$5 \cdot 10^2$	$\mathbb{Z}_8$	0.996	$5 \cdot 10^2$	$\mathbb{F}_8$	0.999
$5 \cdot 10^2$	$\mathbb{Z}_{16}$	0.997	$5 \cdot 10^2$	$\mathbb{F}_{16}$	1.000
$5 \cdot 10^3$	$\mathbb{Z}_4$	1.000	$5 \cdot 10^3$	$\mathbb{F}_4$	1.000
$5 \cdot 10^3$	$\mathbb{Z}_8$	1.000	$5 \cdot 10^3$	$\mathbb{F}_8$	1.000
$5 \cdot 10^3$	$\mathbb{Z}_{16}$	1.000	$5 \cdot 10^3$	$\mathbb{F}_{16}$	1.000

Table: Success frequency of statistical decoding computed over 1000 samples for different integer residue rings and finite fields $t = 3, w = 9$ and $n = 20$.

Comparison

Setup

- rate = $\frac{1}{2}$ ($k = \frac{n}{2}$);

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- $T'_{\text{SD}} = -$ the precomputation;
- $T_{\text{BJMM}} =$ Complexity of one of the best-known ISD algorithm.

Comparing Decoding Algorithm over $\mathbb{Z}_{2^8}$ (Lee)

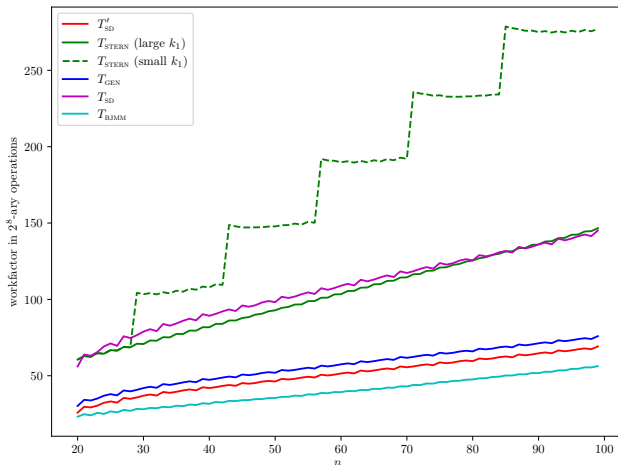


Figure: Workfactors for statistical decoding and Stern algorithms over $(\mathbb{Z}_{2^8})^n$ embedded with the Lee metric.

Comparing Decoding Algorithm over $\mathbb{Z}_{2^9}$ (Lee)

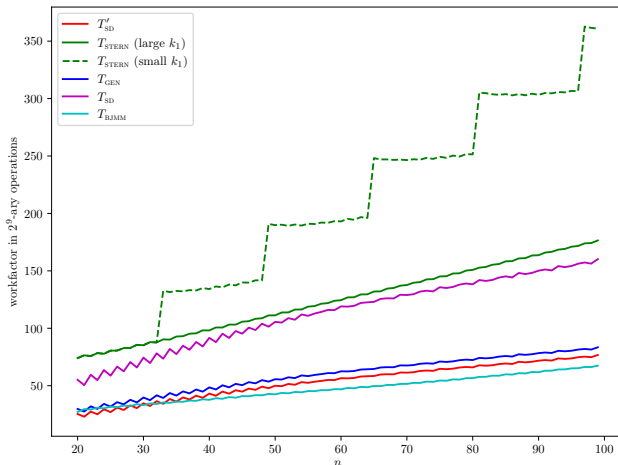


Figure: Workfactors for statistical decoding and Stern algorithms over $(\mathbb{Z}_{2^9})^n$ embedded with the Lee metric.

Niebuhr's Observation holds over the Lee metric

Algorithm \ Ring	$\mathbb{Z}_{2^{10}}$	$\mathbb{Z}_{2^{11}}$	$\mathbb{Z}_{2^{12}}$
BJMM	37.206	43.981	49.833
SD	45.169	36.636	31.547
SD'	20.200	15.974	13.637

Table: Workfactors of statistical decoding and BJMM for $n = 25$ over different rings.

Discussion

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Future Works

Does Niebuhr's observation translate to the Lee metric? Yes but...

- For very large rings;
- Improve the Lee statistical decoding algorithm;
- Overbeck's improvement [9];
- Statistical decoding 2.0 [3];
- Study better algorithm for the computation of $\mathcal{H}_w$.

Questions ?



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